

Intelligent Research Paper Analysis

AI-Driven Optimization of Renewable Energy Scheduling Using Adaptive Genetic Algorithms

1. Paper Overview

Title	AI-Driven Optimization of Renewable Energy Scheduling Using Adaptive Genetic Algorithms
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2. Executive Summary

This paper presents an AI-driven smart optimization framework using an Adaptive Genetic Algorithm (AGA) integrated with a novel Systematic Entropy-Guided Process (SEGP) for multi-objective renewable energy scheduling. The model addresses the intermittency of solar and wind power in a smart grid containing solar, wind, and hydro resources. By adaptively tuning crossover and mutation rates based on population fitness entropy, the method prevents premature convergence and maintains an optimal balance between exploration and exploitation. Experimental evaluation over a 168-hour horizon demonstrates that AGA-SEGP achieves a 96.7% Energy Utilization Rate, a 100% Renewable Contribution Ratio, and a competitive cost of \$0.058 per kWh in just 7.8 seconds of computation.

3. Research Problem

The intermittent and weather-dependent nature of renewable sources (solar and wind) causes supply-demand imbalances, grid instability, curtailment, and costly backup requirements. Conventional deterministic methods struggle with high-dimensional nonlinear scheduling, while standard Genetic Algorithms with fixed mutation and crossover rates suffer from premature convergence and suboptimal solutions due to loss of population diversity.

4. Objectives

- Design a multi-objective renewable energy scheduling model that minimizes operating costs while maximizing renewable energy utilization under dynamic physical and grid constraints.
- Develop and implement the Systematic Entropy-Guided Process (SEGP) within an Adaptive Genetic Algorithm to dynamically regulate crossover and mutation rates based on real-time population diversity.
- Evaluate the proposed algorithm against non-adaptive baseline strategies (Equal Distribution and Cost Priority) on a simulated 168-hour smart grid horizon.

5. Methodology

The methodology is structured into three core components: (1) Problem formulation over a 168-hour (7-day) horizon with hourly resolution, defining a weighted multi-objective fitness function that balances cost minimization and energy utilization maximization subject to power balance, source capacity, and hydro ramp rate constraints; (2) A Genetic Algorithm structure utilizing real-valued chromosome encoding, Binary Tournament Selection, Blend Crossover (BLX- α with $\alpha = 0.5$), and Gaussian mutation; and (3) The Systematic Entropy-Guided Process (SEGP), which calculates population diversity (Div_k) from the standard deviation and mean fitness of generation k to linearly scale crossover rates (Pc) and inversely scale mutation rates (Pm). The system is implemented in Python using NumPy and Pandas.

6. Dataset

Simulated medium-scale smart grid over a 168-hour (one-week) horizon with hourly resolution. Data includes real-world historical hourly load demand (total demand of 41,946 kWh), simulated hourly solar irradiance for PV, simulated hourly wind speed for wind turbines, and dispatchable hydro generation (total available capacity of 75,908 kWh, demand-to-capacity ratio of 0.55).

7. Algorithms & Models

- Adaptive Genetic Algorithm (AGA)
- Systematic Entropy-Guided Process (SEGP)
- Multi-Objective Genetic Algorithm (MOGA)
- Binary Tournament Selection
- Blend Crossover (BLX- α)
- Gaussian Mutation

8. Key Findings

- The SEGP mechanism successfully averted premature convergence by raising mutation rates when diversity dropped and lowering mutation to exploit promising areas when diversity was high.
- Fitness score steadily improved from 0.867 to 1.222 across 100 generations.
- The model achieved an Energy Utilization Rate (EUR) of 96.7%, surpassing the target threshold of 85% by generation 40.
- A 100% Renewable Contribution Ratio was achieved, satisfying all demand without non-renewable backup.
- The optimal energy mix allocated 98.0% of energy to dispatchable Hydro, 1.2% to Solar PV, and 0.8% to Wind Turbines to ensure grid stability and minimize unmet demand penalties.
- AGA-SEGP outperformed the Equal Distribution strategy (60.4% EUR, \$0.053/kWh) and Cost Priority strategy (99.6% EUR, \$0.085/kWh) by delivering a balanced operational cost of \$0.058/kWh with high reliability.

9. Results

The AGA-SEGP achieved a final Energy Utilization Rate of 96.7% (0.967), total operational cost of \$2356.63 (\$2346.15 in baseline comparison), unit cost of \$0.058 per kWh, cumulative unmet energy demand of 1404.3 kWh, 100.0% renewable contribution ratio, and a fast optimization wall-clock runtime of 7.8 seconds across 100 generations with a population size of 100.

10. Limitations

- The resulting dispatch relies heavily on hydro power (98.0% of energy mix) to compensate for intermittency and avoid penalties.
- The framework currently omits energy storage elements such as batteries.
- Generation and demand profiles are handled deterministically without explicit stochastic modeling of forecast error distributions.

11. Research Gap

Standard evolutionary algorithms for smart grid scheduling rely on static crossover and mutation parameters, leading to loss of diversity and suboptimal local minima when balancing conflicting multi-objective constraints

under intermittent generation.

12. Future Scope

- Incorporate optimization for Battery Energy Storage Systems (BESS), including charge/discharge cycles, battery degradation constraints, and fluctuating electricity market prices.
- Integrate stochastic modeling (e.g., Monte Carlo simulations) to explicitly account for forecast uncertainties and error margins in solar and wind generation.
- Apply the SEGP adaptive principle to other complex smart grid optimization problems, such as Optimal Power Flow (OPF) and distribution network reconfiguration.

13. Key Concepts

- Adaptive Genetic Algorithm (AGA)
- Systematic Entropy-Guided Process (SEGP)
- Population Diversity / Information Entropy
- Renewable Energy Scheduling
- Smart Grid Energy Management
- Multi-Objective Optimization
- Energy Utilization Rate (EUR)
- Renewable Contribution Ratio (RCR)
- Hydro Ramp Rate Limits
- Blend Crossover (BLX- α)

14. Overall Evaluation

The paper provides a well-formulated, lightweight, and effective adaptive evolutionary computation framework for multi-source renewable energy scheduling. By introducing an entropy-guided mechanism to dynamically control genetic operators, it resolves the premature convergence drawback of standard GAs, yielding high utilization, low operational costs, and near-instant computational execution suitable for practical smart grid dispatch.